I checked your uploaded Python file, attendence recovery.py. I'll prepare the complete activity record based on your actual code, in simple language suitable for a B.Tech 1st-year CSE Data Science student.

Your program calculates the current attendance percentage and finds how many additional classes a student must attend consecutively to reach the 75% attendance target.

Attendance Recovery Calculator

1. Problem statement

To develop a Python program that calculates a student's current attendance percentage and determines the number of additional classes the student must attend to achieve the required 75% attendance. The program should also validate the attendance details and display the results.

2. Objectives

1. To calculate the current attendance percentage.
2. To check whether the entered attendance details are valid.
3. To determine the number of additional classes required to reach 75% attendance.
4. To understand the use of if-else statements and while loops.
5. To practice variables, arithmetic operations, and input/output in Python.

3. Assumptions

- The number of classes attended and total classes held are integers.
- Total classes held must be greater than zero.
- Classes attended cannot be negative or greater than total classes held.
- The required attendance percentage is fixed at 75%.
- Every additional class is attended by the student.
- The program calculates the required classes without considering future absences.

4. Algorithm

1. Start the program.
2. Read the number of classes attended.
3. Read the total number of classes held.
4. Set the target attendance to 75%.
5. Check whether the attendance details are valid.
6. If the details are invalid, display an error message.
7. Otherwise, calculate the current attendance percentage.
8. Initialize the additional classes counter to zero.
9. Use a while loop to increase the counter until the attendance percentage reaches 75%.

10. Display the current attendance percentage, additional classes required, and target attendance.
11. Stop the program.

5. Flowchart

Start

Input attended classes and total classes

Are attendance details valid?

Yes

Calculate attendance percentage

Set extra classes = 0

Has attendance reached 75%?

If no, increase extra classes and repeat the check

Display results

No

Display invalid attendance message

End

End

Note: The flowchart is a simplified representation of the program. The loop repeats until the target is reached.

6. Pseudocode

```
BEGIN
   INPUT attended
   INPUT total
   target = 75

   IF total <= 0 OR attended < 0 OR attended > total
      DISPLAY "Invalid attendance details."
   ELSE
      percentage = attended / total * 100
      extra = 0

      WHILE (attended + extra) /
           (total + extra) * 100 < target
```

```
        extra = extra + 1
    END WHILE


    DISPLAY current attendance percentage
    DISPLAY extra classes required
    DISPLAY target attendance
  END IF
END
```

7. Python source code

The following code follows your uploaded program's logic.

```
attended = int(input("Classes attended: "))total = int(input("Total classes held: "))target = 75if total <= 0 or attended < 0 or attended > total:    print("Invalid attendance details.")else: percentage = attended / total * 100    extra = 0    while (attended + extra) / (total + extra) * 100 < target:        extra += 1    print("Current attendance:", round(percentage, 2), "%") print("More classes to attend:", extra)    print("Target attendance:", target, "%")
```

8. Test cases

Test case	Classes attended	Total classes	Expected result
1	60	100	60% attendance; attend 60 more classes
2	70	100	70% attendance; attend 20 more classes
3	75	100	75% attendance; 0 more classes
4	0	0	Invalid attendance details
5	110	100	Invalid attendance details
6	40	80	50% attendance; attend 80 more classes

9. Output screenshots

The following are sample output previews. They are not actual screenshots captured from a Python interpreter.

Test 1 — Attendance below target

```
Classes attended: 60
Total classes held: 100
Current attendance: 60.0 %
More classes to attend: 60
Target attendance: 75 %
```

Copy output

Test 2 — Attendance already at target

Classes attended: 75
Total classes held: 100
Current attendance: 75.0 %
More classes to attend: 0
Target attendance: 75 %

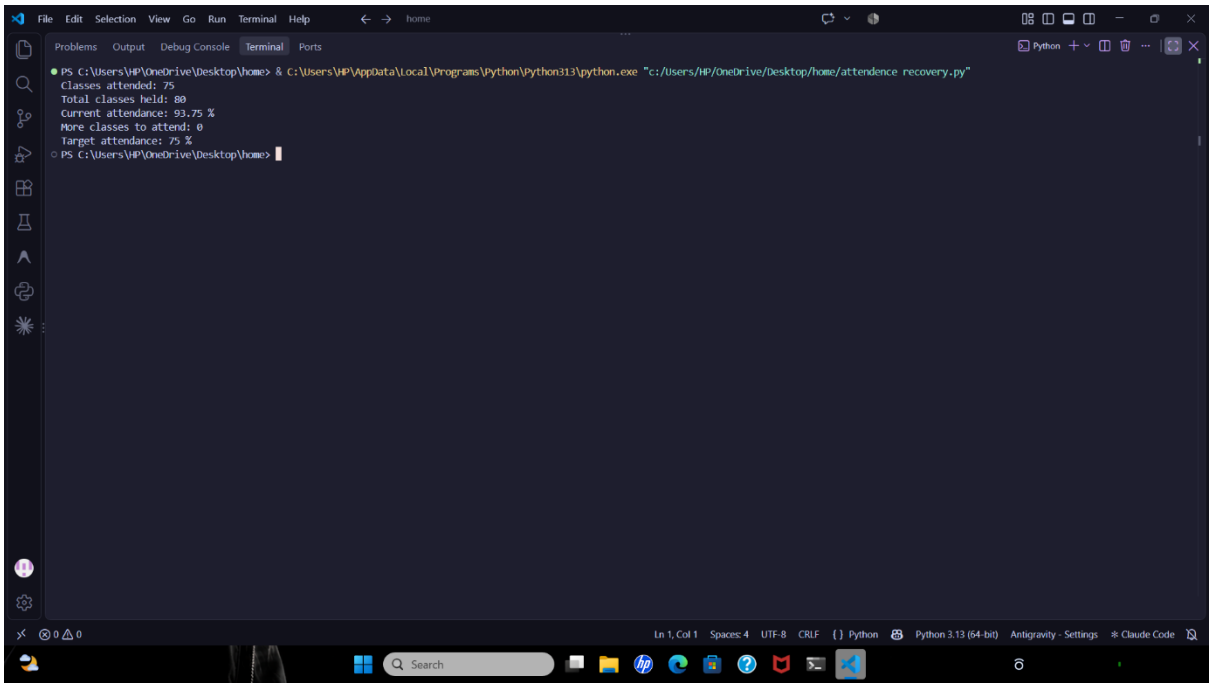
Copy output

Test 3 — Invalid attendance details

Classes attended: 110
Total classes held: 100
Invalid attendance details.

Copy output

To include actual screenshots in your activity record, run the program in Python IDLE or VS Code, enter the test values, and capture the console output.



10. Conclusion

The Attendance Recovery Calculator was successfully developed using Python. It calculates the student's current attendance percentage and determines how many additional classes must be attended to reach the 75% target. It also validates attendance details and displays an error message for invalid inputs. This program helps us understand conditional statements, while loops, variables, and arithmetic operations in Python.